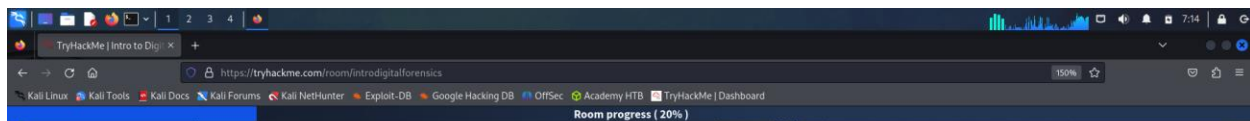


1. Introduction

Delving into digital forensics reveals its crucial role in modern crime investigation. With the proliferation of digital systems, specialized forensic techniques are essential. Law enforcement encountering digital devices at crime scenes prompts questions about effective evidence collection, transfer, and analysis. Each device potentially holds vital information, highlighting the importance of digital forensics. This discipline aids both public and private sectors in unraveling crimes or policy violations. Skilled digital forensics investigators are vital for preserving evidence integrity. Understanding digital forensics intricacies is essential for justice and cybersecurity.

2. Module Sections



✓ Woop woop! Your answer is correct

Answer the questions below

Consider the desk in the photo above. In addition to the smartphone, camera, and SD cards, what would be interesting for digital forensics?

laptop

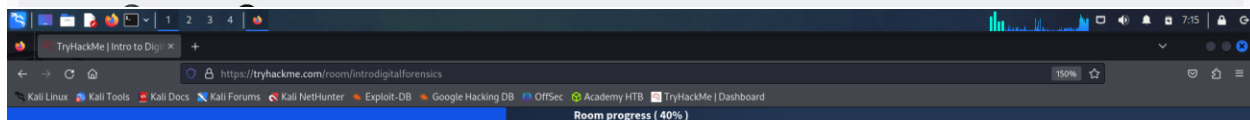
✓ Correct Answer

🔍 Hint

Task 2 ○ Digital Forensics Process

Task 3 ○ Practical Example of Digital Forensics

Created by	Room Type	Users in Room	Created
------------	-----------	---------------	---------



More generally, according to the former director of the Defense Computer Forensics Laboratory, Ken Zatyko, digital forensics involves several key principles:

✓ Woop woop! Your answer is correct

- Proper search authority: Investigators cannot commence without the proper legal authority.
- Chain of custody: This is necessary to keep track of who was holding the evidence at any time.
- Validation with mathematics: Using a special kind of mathematical function, called a hash function, we can confirm that a file has not been modified.
- Use of validated tools: The tools used in digital forensics should be validated to ensure that they work correctly. For example, if you are creating an image of a disk, you want to ensure that the forensic image is identical to the data on the disk.
- Repeatability: The findings of digital forensics can be reproduced as long as the proper skills and tools are available.
- Reporting: The digital forensics investigation is concluded with a report that shows the evidence related to the case that was discovered.

Answer the questions below

It is essential to keep track of who is handling it at any point in time to ensure that evidence is admissible in the court of law. What is the name of the documentation that would help establish that?

Chain of Custody

✓ Correct Answer

Task 3 ○ Practical Example of Digital Forensics

To find the author's name I ran pdftinfo against the pdf and found the author as shown below

The image shows a Kali Linux desktop environment. In the foreground, a web browser displays a challenge page from TryHackMe. The page shows the PDF metadata for 'ransom-letter.pdf', indicating it was created by Microsoft Word 2016 on October 10, 2018. A text input field contains the author's name, 'Ann Gree Shepherd', and a green button indicates the answer is correct. Below this, the 'Photo EXIF Data' section is partially visible.

In the background, a terminal window shows the following commands and output:

```
root@ip-10-10-26-161:~/Rooms# ls
ADenumeration  CVE2022-26923  ParrotPost
Aoc2023        ECB             PKI
Aoc2024        ExploitingAD   securityfootage
Aoc3           Follina-MSDT  sigma
BPVolatility   InsecureRandomness  solar
BreachingAD    Introdigitalforensics  SplunkBasic
caldera        introloganalysis  Spring4Shell
CapstoneChallenge  JohnTheRipper  Tomcat-CVE-2024-50379
cauldron       LengthExtensionAttacks  TryHack3M
Compiled        loganalysis      TryPwnMeOne
cryptographyintro  logs            TryPwnMeTwo
ctf-event-2022  monikerlink     Wireshark101
CVE2022-26134  OhSINT

root@ip-10-10-26-161:~/Rooms# cd introdigitalforensics/
root@ip-10-10-26-161:~/Rooms/introdigitalforensics# ls
letter-image.jpg  ransom-letter.pdf
ransom-letter.doc  ransom-letter-2.zip

root@ip-10-10-26-161:~/Rooms/introdigitalforensics# pdftinfo ransom-letter.pdf
Title:      Pay NOW
Subject:     We Have Gato
Author:      Ann Gree Shepherd
Creator:     Microsoft® Word 2016
Producer:    Microsoft® Word 2016
CreationDate: Wed Feb 23 09:10:36 2022 GMT
ModDate:     Wed Feb 23 09:10:36 2022 GMT
Tagged:      yes
UserProperties: no
Suspects:    no
Form:        none
JavaScript:  no
Pages:       1
Encrypted:    no
Page size:   595.44 x 842.04 pts (A4)
Page rot:    0
File size:   71371 bytes
Optimized:   no
PDF version: 1.7

root@ip-10-10-26-161:~/Rooms/introdigitalforensics#
```

I ran exiftool against the letter.jpg file and found the GPS coordinates of where the image was taken. I converted the coordinates to the format shown and searched on Bing maps and found the exact street where the image was taken from as shown below.

TryHackMe | Intro to Digi... THM Browser-Based GPS coordinates, latitud...
https://www.gps-coordinates.net
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard

login | register

Coordinates My Location Driving Directions Converter US Map Satellite Street View API Maps Distance

Latitude 51.5144167
Longitude -0.09409166666666666
Get Address

Lat,Long
DMS (degrees, minutes, seconds)*
Latitude N O S 51 30 51.90
Longitude E W 0 5 38.73
Get Address

* World Geodetic System 84 (WGS 84)

Advertisement: BURBERRY HERO

TryHackMe | Intro to Digi... THM Browser-Based Bing Maps - Directions...
https://www.bing.com/maps?cp=51.514865~-0.093105&lv=19.0
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard

Microsoft Bing 51.514417, -0.094092

Directions Traffic My Places More

8 Russia Row, EC2V 8BL, England, United Kingdom
Directions Nearby
Save to calendar

Nearby
Parking Restaurants Gas stations
Hotels Coffee Nearby Transit
Search nearby

United Kingdom - England - Greater London - London
51.514416, -0.094092

Map showing streets: Wood Street, Milk Street, Gresham Street, King Street, Trump Street, Lawrence Lane, Ironmonger Lane, Russia Row. Landmarks: Guildhall Buildings, London Persian Rug Company, SOHO Coffee, City of London, The Old Church of Saint Lawrence Jewry, Commerzbank, Pret A Manger, PT Bank (Persero), Star Bank of India, Qube Global Software.

Privacy and Cookies Legal Advertise Help Feedback
© 2025 Microsoft

TryHackMe | Intro to Digital Forensics

THM Browser-Based

Bing Maps - Directions

https://tryhackme.com/room/introdigitalforensics

Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard

Room progress (60%)

user@TryHackMe\$ exiftool IMAGE.jpg

[...]

GPS Position : 51 deg 31' 4.00" N, 0 deg 5' 48.30" W

[...]

If you take the above coordinates and search one of the online maps, you will learn more about this location. Searching [Microsoft Bing Maps](#) or [Google Maps](#) for **51 deg 30' 51.90" N, 0 deg 5' 38.73" W** reveals the street where the photo was taken. Note that for the search to work, we had to replace **deg** with **°** and remove the extra white space. In other words, we typed **51°30'51.9"N 0°05'38.7"W** in the map search bar.

Using **exiftool** or any similar tool, try to find where the kidnappers took the image they attached to their document. What is the name of the street?

milk street

✓ Correct Answer

🔑 Hint

What is the model name of the camera used to take this photo?

Applications Places System Tue 20 May, 12:42

Woop woop! Your answer is correct

File Edit View

Chromaticity Channel 3 : 0.15001 0.06

Device Mfg Desc : GIMP

Device Model Desc : sRGB

Current IPTC Digest : b417d6571f8aba97a1e64afbdedafbdb

Coded Character Set : UTF8

Envelope Record Version : 4

Date Created : 2022:02:15

Digital Creation Date : 2021:11:05

Digital Creation Time : 14:06:13+03:00

Application Record Version : 4

Time Created : 17:23:40-17:23

Image Width : 1200

Image Height : 800

Encoding Process : Progressive DCT, Huffman coding

Bits Per Sample : 8

Color Components : 3

Y Cb Cr Sub Sampling : YCbCr4:4:4 (1 1)

Aperture : 2.8

Image Size : 1200x800

Megapixels : 0.960

Scale Factor To 35 mm Equivalent: 0.7

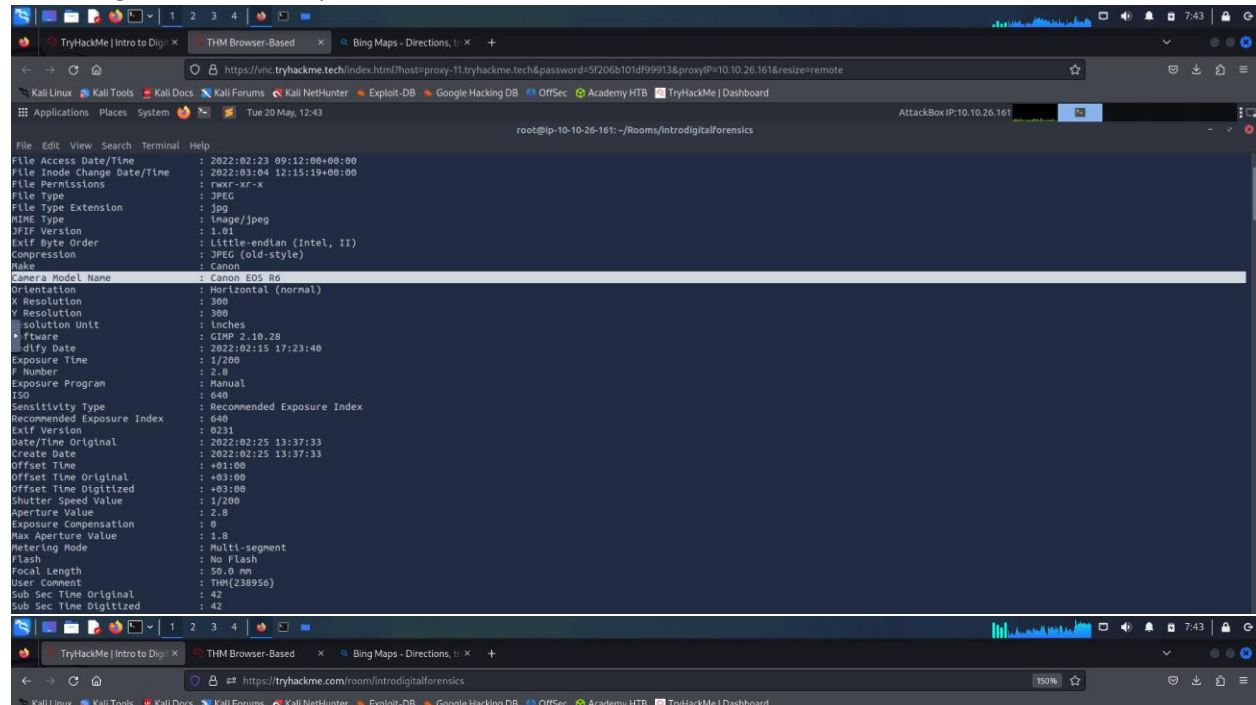
Shutter Speed : 1/200

Create Date : 2022:02:25 13:37:33.42+03:00

THM AttackBox

39min 54s

Searching the exiftool output I discovered the camera's model name to be as shown below.



```
File Access Date/Time       : 2022:02:23 09:12:00+00:00
File Inode Change Date/Time : 2022:03:04 12:15:19+00:00
File Permissions           : rw-r--r--x
File Type                  : JPEG
File Type Extension        : jpg
MIME Type                  : image/jpeg
JFIF Version               : 1.01
Exif Byte Order            : Little-endian (Intel, II)
Compression                : JPEG (old-style)
Make                       : Canon
Camera Model Name          : Canon EOS R6
Orientation                 : Horizontal (normal)
X Resolution               : 300
Y Resolution               : 300
Resolution Unit             : inches
Software                   : GIMP 2.10.28
Modify Date                : 2022:02:15 17:23:40
Exposure Time              : 1/200
F Number                   : 2.8
Exposure Program           : Manual
ISO                        : 640
Sensitivity Type           : Recommended Exposure Index
Recommended Exposure Index : 640
Exif Version               : 0231
Date/Time Original         : 2022:02:25 13:37:33
Create Date                : 2022:02:25 13:37:33
Offset Time                : +01:00
Offset Time Original       : +03:00
Offset Time Digitized      : +03:00
Shutter Speed Value        : 1/200
Aperture Value             : 2.8
Exposure Compensation      : 0
Max Aperture Value         : 1.8
Metering Mode              : Multi-segment
Flash                     : No Flash
Focal Length               : 50.0 mm
User Comment               : THM(238956)
Sub Sec Time Original      : 42
Sub Sec Time Digitized     : 42
```

Woop woop! Your answer is correct

Congratulations on completing Intro to Digital Forensics!!! 🎉

Points earned 🔥 40	Completed tasks 📋 3	Room type 👤 Walkthrough	Difficulty 📶 Easy	Streak 🔥 1
-----------------------	------------------------	----------------------------	----------------------	---------------

Leave Feedback

Next

4. Social Media Posting

https://x.com/intent/post?url=https%3A%2F%2Ftryhackme.com%2Froom%2Fintrodigitalforensics%3Futm_source%3Dtwitter%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom&text=Share%20your%20achievement&via=realtryhackme&hashtags=tryhackme

https://www.facebook.com/sharer/sharer.php?u=https%3A%2F%2Ftryhackme.com%2Froom%2Fintrodigitalforensics%3Futm_source%3Dfacebook%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom&t=Share%20your%20achievement

https://www.linkedin.com/sharing/share-offsite/?url=https%3A%2F%2Ftryhackme.com%2Froom%2Fintroduction-to-digital-forensics%3Futm_source%3Dlinkedin%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom

5. Conclusion

In wrapping up, our journey through digital forensics has emphasized its pivotal role in modern crime investigation. With the prevalence of digital systems, specialized techniques become crucial for unraveling complex cases. Reflecting on scenarios where law enforcement encounters digital devices, I acknowledge the critical questions surrounding evidence handling. Proficiency in digital forensics practices is paramount for uncovering crucial information, whether in public or private sectors. As an investigator, my role in preserving evidence integrity cannot be overstated. Embracing a deeper understanding of digital forensics is essential for upholding justice and bolstering cybersecurity efforts.